TECTONICS, VOL. 9, NO. 4, PAGES 719-734, AUGUST 1990

TECTONIC HISTORY OF THE LUNKSOOS COMPOSITE TERRANE IN THE MAINE APPALACHIANS

M. Reid Wellensiek, Ben A. van der Pluijm, Rob Van der Voo, and Rex J. E. Johnson

Department of Geological Sciences, University of Michigan, Ann Arbor

Abstract. Several Ordovician volcanic terranes are preserved between the Paleozoic North American margin and the Avalon Terrane in the northern Appalachians of Maine. They have been interpreted as remnants of an island arc on the basis of their current position in the Appalachians or as ocean islands near the margin of Avalon on the basis of paleontologic data. A study of one of these terranes, the Lunksoos terrane in northern Maine, was undertaken through a combination of paleomagnetic and structural analyses in order to understand its tectonic evolution. Paleomagnetic study of the pre-Caradocian (pre-Late Ordovician) Stacyville volcanics revealed three components of magnetization. The final magnetization removed (D=90°, I=36°, $\alpha 95=11.8^{\circ}$) is interpreted to be primary and yields a paleolatitude of approximately 20°S; however, a Late Ordovician remagnetization cannot be excluded. The Stacyville volcanics now overlie a Cambrian and Lower Ordovician sequence (the Grand Pitch and Shin Brook formations) that is believed to have been deposited at the southern side of Iapetus, possibly near the margin of Avalon. During the Middle and Late Ordovician, Avalon was at paleolatitudes higher than 40°S, while the volcanics were either formed or remagnetized at a paleolatitude of 20°S. The pre-Caradocian Stacyville volcanics therefore could not have been deposited directly on the Grand Pitch-Shin Brook sequence but must have been tectonically emplaced. The Rockabema Diorite which intrudes the entire Cambro-Ordovician sequence constrains the timing of this emplacement to be pre-late Early Silurian. These relationships indicate that the Lunksoos area is a composite terrane within the Maine Appalachians. The tectonic emplacement scenario is unnecessary if the Grand Pitch and Shin Brook formations were not located on the south side of Iapetus during the Ordovician. This alternative, however, poses conflicts with the interpretation of faunal affinities.

Copyright 1990
by the American Geophysical Union.

Paper number 89TC03077.
0278-7407/90/89TC-03077$10.00

INTRODUCTION

Since the recognition of major tectonostratigraphic zones or terranes in the northern Appalachians [Williams, 1979; Williams and Hatcher, 1983], attention has been focused on defining smaller terranes within these zones [Zen, 1983; Keppie, 1985]. In Maine there are several distinct volcanic terranes that contain non-North American faunas [Neuman, 1984]. These terranes now lie between the Paleozoic North American margin and the Avalon Terrane. Using the subdivisions of Newfoundland and New Brunswick, this area may be considered to be part of the intervening Central Mobile Belt [van der Pluijm and van Staal, 1988]. Paleontologic evidence suggests that these terranes were located at some distance from the North American margin; several may have been, in fact, closer to the Avalonian margin [Neuman, 1984]. The latitudinal separation between North America and these terranes, however, can rarely be estimated from faunal evidence alone. In addition, the timing and nature of accretion of these terranes is not well known. The intent of this study

was to undertake a detailed analysis of one of these volcanic terranes in order to constrain its paleogeographic position during the Ordovician and study its tectonic evolution. The terrane chosen was the Lunksoos terrane in northern Maine (Figure 1).

The terrane analysis was approached through a combination of field work and paleomagnetic study. Detailed structural mapping was instrumental in identifying the timing and extent of deformation within the terrane. Paleomagnetic data from the sampled sequence of Middle (?) Ordovician volcanics allowed the paleolatitude at the time the volcanics were magnetized to be calculated and thus paleolatitudinal separations could be determined.

GEOLOGIC SETTING

Within Maine three major zones have been identified by isotopic, structural, and stratigraphic studies [e.g., Osberg, 1978; Zen, 1983; Ayuso, 1986; Ludman, 1986; Zartman, 1988]. Although different names have been used to describe these zones, there is general agreement on their characteristics. The westernmost zone, the Connecticut-Gaspe Synclinorium, lies to the west of the interpreted autochthonous Ordovician margin of North America [Zen, 1983]. The easternmost zone corresponds to the Avalon Terrane [Zartman, 1988], with a distinct stratigraphy [Zen, 1983] and lead isotopic signature of Devonian granites [Ayuso, 1986]. The intervening area, the Kearsarge-Central Maine Synclinorium [Zartman, 1988] or the Merrimack Synclinorium [Zen, 1983], contains the Lunksoos terrane and other Ordovician volcanic terranes. Prior to the Acadian Orogeny, the position of the Kearsarge-Central Maine Synclinorium relative to North America is not known [Zartman, 1988]. The lead isotopic ratios give intermediate values compared to the zones to the east and west and indicate an imprint of a continental crustal source [Ayuso, 1986]. Thus the zone appears to have been a distinct crustal fragment at some point in geologic time. This central zone has been considered part of an island arc in several plate tectonic models [Bird and Dewey, 1970; Osberg, 1978], although the polarity of subduction is uncertain.

The Lunksoos area was mapped by R.B. Neuman and described in his report [Neuman, 1967]; our mapping focused on structural relationships. Within the terrane a sequence of Paleozoic rocks is preserved in a series of anticlines and synclines. In the south the terrane is interrupted by the intrusion of the Devonian Katahdin pluton [Rankin and Hon, 1987], while in the north it is buried beneath Silurian and Devonian cover. The oldest unit exposed in the area is the Grand Pitch Formation (Figure 1). The sediments of this formation consist of red, green, and gray slate, siltstone, and quartzite, which commonly occur as graded beds. The Grand Pitch Formation is assigned an Early Cambrian (?) age on the basis of the presence of the trace fossil *Oldhamia smithi* found in finely laminated red slates and siltstones [Smith, 1928; Neuman, 1962]. *Oldhamia* cannot be assigned to a specific biogeographic province since it has been found in widely separated localities throughout the world [e.g., Lindholm and Casey, 1989]. Overlying the Grand Pitch Formation is the Shin Brook Formation, which consists of tuffs with minor interbedded flows, tuffaceous sandstone, and breccia (Figure 1). The Shin Brook Formation is preserved in a doubly plunging, northeast-southwest trending syncline. Brachiopods found within the sandstone layers are Arenigian (Early Ordovician) in age [Neuman, 1964]. They are associated with the Celtic faunal province and therefore have a non-North American affinity [Neuman, 1984].

In the southern end of the terrane a thick sequence of unnamed volcanics are exposed (Figure 1). We will informally refer to these volcanics as the Stacyville volcanics. The Stacyville volcanics, which were sampled for paleomagnetic study, consist of fine-grained gray to greenish-gray pillow lavas and minor interbedded flows and flow breccia that have an estimated total thickness of about 2700 m [Neuman, 1967]. Farther to the north, the volcanics are absent. Hynes [1976, 1981] carried out whole rock major and trace element and pyroxene major element analyses on the Stacyville volcanics and concluded that the volcanics were formed in an island arc setting. Along the Wassataquoik Stream, volcanics and cherts are interlayered at the boundary between the Stacyville volcanics and the overlying Wassataquiok Chert. The chert above the transition is thinly bedded, medium to dark gray with alternating bands of light gray. Graptolites and conodonts of Caradocian (early Late Ordovician) age have been found in the Wassataquoik Chert [Neuman, 1984]. Thus the age of the volcanics is constrained by the overlying cherts to be pre-Late Ordovician (pre-Caradocian). No maximum age can be determined for the volcanics since the lower contacts are unlikely to be conformable, as will be discussed later. However, considering the upper age limit, the thickness of the sequence, and regional relationships, a Middle Ordovician age is most likely.

Within the terrane the Stacyville volcanics are overlain by two distinct Late Ordovician units (Figure 1). In the southeast, as previously discussed, the Wassataquoik Chert is exposed. In the northwest, near Pond Pitch, the Late Ordovician sequence consists of sandstone, siltstone, conglomerate, and some interbedded volcanics. Brachiopods in the sandstone and the matrix of the conglomerate suggest an Ashgillian age for these units [Neuman, 1967]. The mixed Northern European and North American faunal affinities of the brachiopods do not yield conclusive information about the provenance of this unit, however [Neuman, 1984]. By the Late Ordovician the faunal differences of the opposite sides of the Iapetus Ocean were not very pronounced [Cocks and Fortey, 1982], indicating that the ocean had narrowed

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

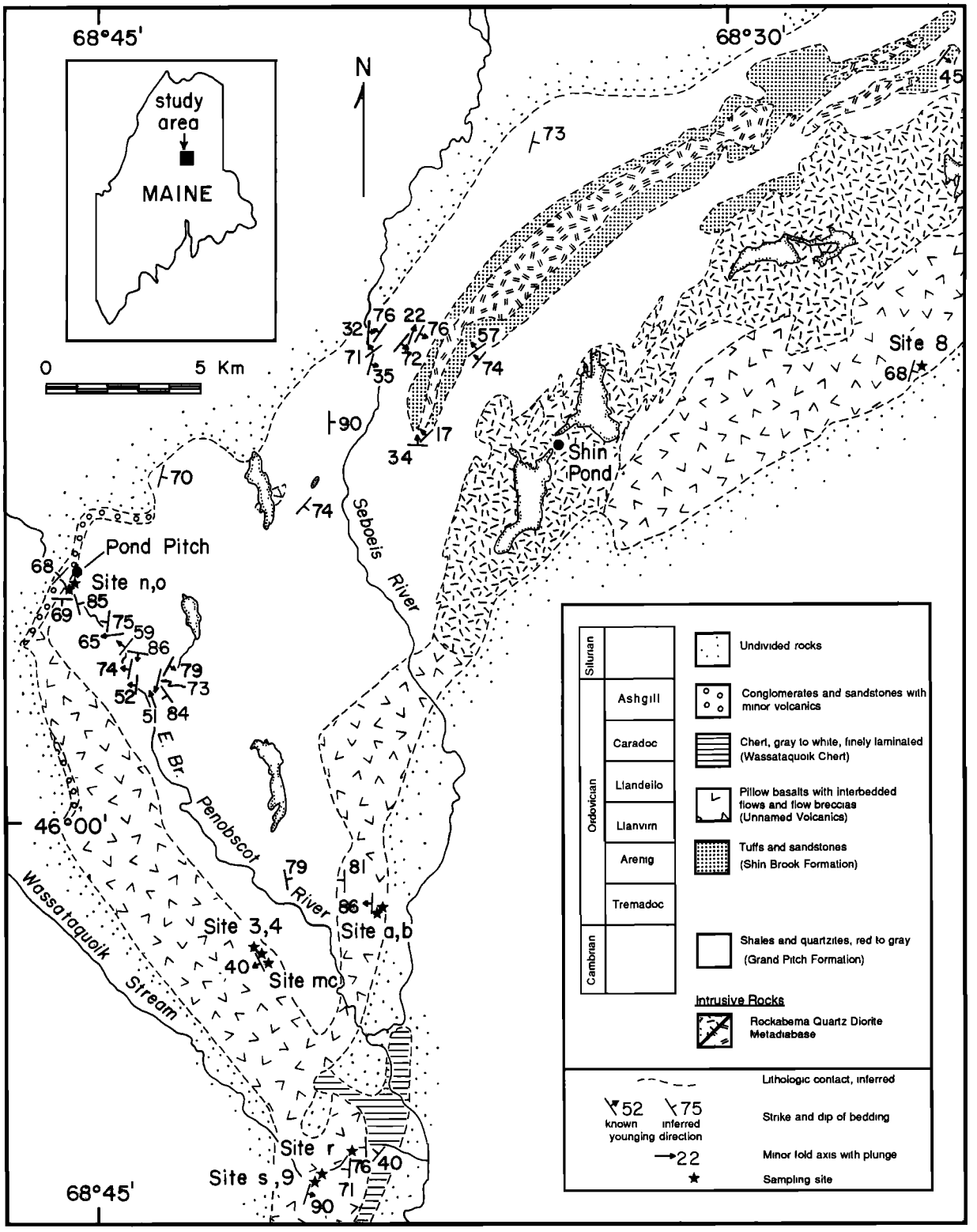


Fig. 1. Lithologic map of the study area in northern Maine showing distribution of rock units and sampling localities. Labels for the sites have been abbreviated to the last letter or number of the site name, (i.e., site wva is site a on map).

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

enough so that Avalon was within dispersal range of North American benthic habitats.

The Cambro-Ordovician sequence is intruded by the Rockabema Diorite (Figure 1). The intrusive nature of the formation is seen where it is in contact with the Grand Pitch Formation and the Stacyville volcanics. The timing of the intrusion of the diorite is constrained to be older than Early Silurian, because clasts of the diorite are found in a late Llandoverian (Early Silurian) conglomerate [Neuman, 1967]. Additionally, it must of course be younger than the pre-Caradocian Stacyville volcanics which it intrudes.

Lithologic contacts within the terrane are rarely exposed, and where they are, it is generally difficult to distinguish between conformable, unconformable, and faulted contacts. The Grand Pitch Formation, for example, is overlain by units ranging from Early Ordovician to Silurian in various parts of the terrane. In the north central area the Grand Pitch Formation is overlain by the Early Ordovician Shin Brook Formation. This contact was described as an unconformity on the basis of a conglomerate at the base of the Shin Brook Formation and the apparently more complex structure in the Grand Pitch Formation [Neuman, 1967]. The conglomerate has clasts of slate and quartzite, presumably derived from the Grand Pitch Formation. The presence of this basal conglomerate has also been cited as evidence of the Penobscot deformation event [Neuman, 1967]. It is not clear, however, whether this conglomeratic base is the result of a true angular unconformity or simply a change in depositional environment. On the western edge of the terrane, near Pond Pitch, the Middle (?) Ordovician Stacyville volcanics overlie the Cambrian Grand Pitch Formation. These volcanics are in turn overlain by Ashgillian conglomerates, clastics, and volcanics. The Ashgillian sequence, however, directly overlies the Cambrian Grand Pitch Formation to the north of Pond Pitch, and the intervening Stacyville volcanics are absent. The Caradocian Wassataquoik Chert previously described as overlying the volcanics in the south is absent in this area. Farther to the northeast, the Silurian cover sequence directly overlies the Grand Pitch, and all intervening Ordovician units are absent. Throughout the area, therefore, the contacts between the Cambrian Grand Pitch Formation and the overlying units are either disconformable or faulted.

METAMORPHIC AND DEFORMATION HISTORY

The metamorphic grade of the Lunksoos area is subgreenschist to lower-greenschist facies, and the peak of metamorphism is considered Devonian in age [Neuman, 1967; Osberg, 1978]. The relatively low grade of this area makes it suitable for paleomagnetic study. Although thermoviscous remagnetization is a concern in paleomagnetic analyses, premetamorphic magnetizations have been preserved in rocks of similar grade elsewhere in the Appalachians [e.g., Johnson and Van der Voo, 1985].

We have identified two generations of deformation in the area through structural mapping (Figure 1). The first folding event, F1, seems to be localized and is characterized by generally steeply plunging fold axes (Figure 2). There is no mesoscopic cleavage associated with F1 folds. The younger F2 folds are larger-scale features than F1 and are pervasive throughout the map area. The F2 folds are doubly plunging with fold axes that are shallow to subhorizontal and trend roughly N-NNE (Figure 2). A

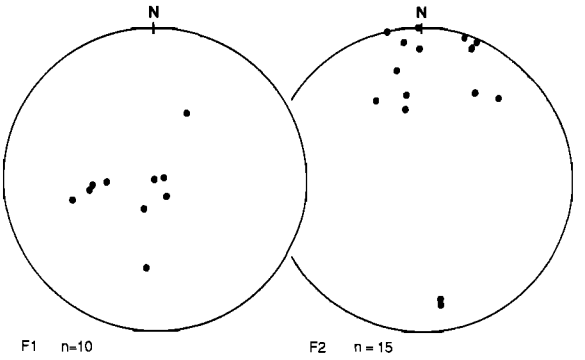


Fig. 2. Lower hemisphere, equal-area projections of F1 and F2 fold axes. Lineations and bedding-cleavage intersections have not been included.

well-developed regional cleavage, S2, is axial planar to the F2 folds and in places is seen to crosscut the F1 folds. No superimposed folding relationships between F1 and F2 folds are seen in the field other than S2 cleavage overprinting. F2 folding is the dominant regional folding event that is responsible for the current steep dips of bedding (Figure 3).

F1 folds are evident in the Cambrian Grand Pitch Formation (see also Neuman [1967]). F1 structures were not found in either the overlying Shin Brook Formation or the Stacyville volcanics, but they are present in the Caradocian Wassataquoik Chert. The absence of F1 structures in the volcanic units is perhaps due to their more massive nature. The age of the F1 folding must postdate the deposition of the cherts and therefore must be post-Caradocian. Since there is no reported F1 deformation in the Siluro-Devonian sequence, F1 folding is most likely Late Ordovician or Early Silurian in age. In contrast, evidence of F2 folding can be seen throughout the entire Cambrian to early Devonian sequence (Figure 3) (see also Neuman [1967]) and is therefore of at least Early Devonian age. This second deformation event coincides with what is generally considered to be the Acadian Orogeny [Rodgers, 1967; Williams and Hatcher, 1983].

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Fig. 3. F2 fold with subhorizontal fold axis in the Grand Pitch Formation, west bank of the Seboeis River at the Grand Pitch.

PALEOMAGNETISM

Sampling and Results

Paleomagnetic samples were collected in both east and west facing successions of the Stacyville volcanics so that a fold test could be applied. Pillow lavas were sampled at sites wva, wvs, wm9, wmc, wv3, and wv4, yielding a total of 52 samples. An additional 43 samples were collected from nonpillow flows at sites wvb, wm8, wvn, wvo, and wvr. In total, eleven sites were sampled (Figure 1). Preliminary geochemical data from the paleomagnetic sites indicate that all the volcanics sampled have similar geochemical patterns [Wellensiek, 1988]. Most of the samples were drilled in the field with a portable gasoline-powered drill and oriented with a magnetic compass and inclinometer. Where drilling was not feasible, oriented hand samples were collected and cores were later drilled in the laboratory.

The natural remanent magnetizations (NRMs) of the samples have intensities that range from 10^{-6} to 10^{-1} A/m. Measurements were made either on a Superconducting Technology cryogenic magnetometer or a Schonstedt SSM-1A spinner magnetometer, depending on the intensity of the sample. All samples were stored and treated in a magnetic field-free room to reduce the effects of viscous magnetizations. Both thermal and alternating field (AF) demagnetizations were performed on a pilot collection of samples. AF demagnetization proved to be less useful since it did not separate the components of magnetization as completely as thermal demagnetization. Furthermore, a more complete decay of the NRM could be achieved with thermal demagnetization. The remainder of the samples were therefore thermally demagnetized in a stepwise fashion.

The demagnetization results were plotted in orthogonal vector diagrams [Zijderveld, 1967] for visual inspection, and magnetic directions of linear trajectories were determined by principal component analysis [Kirschvink, 1980]. Great circle analysis was used where two or more overlapping components of magnetization yielded curvilinear demagnetization trajectories [Halls, 1976]. In three sites (wvn, wvo, and wvr) the remanent magnetizations were unstable, and no analysis could be applied. In site wv3 the magnetizations were stable, but the intensities were anomalously high, and the magnetic directions were random. Since this site is on a hilltop, the possibility of a lightning strike is likely. These four sites were therefore not included in the calculation of mean directions.

Tilt corrections were applied to the in situ directions on the basis of the strike and dip of the volcanics. Most samples were taken from pillows

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

where tops could be easily seen, and thus an accurate strike and dip could be measured for each sample. As discussed above, the F1 folding occurs on a smaller, localized scale and is not found in the volcanics. In addition, the presently observed relationship of steep F1 axes and subhorizontal F2 axes requires that F1 structures originally were recumbent. Consequently, F1 deformation did not result in appreciable tilt of the volcanics. Therefore we only need to correct for the F2 folding, since it is responsible for the current attitude of the volcanics.

Three components of magnetization can be recognized in the samples. A soft, low blocking temperature component is the direction first removed (Figure 4). This direction, the C component, is typically removed by 300°C. The in situ site mean directions are northerly and steep, corresponding to the present-day dipole field direction. Site means for the C component of magnetization fail the fold test, indicating that it is a postfolding magnetization. Thus it is likely to be a secondary viscous magnetization related to the Earth's recent magnetic field.

The second component of magnetization, the B component, is removed between 300°C and 585°C. The in situ direction is southerly with shallow upward and downward inclinations (Figure 4). In many samples this component decays linearly to the origin, and in three sites (wvb, wm8, and wv4) component B can be determined by principal component analysis. In several samples from other sites, however, the trajectory bypasses the origin, indicating that a third component of magnetization remains. In sites wmc and wva the blocking temperature of component B overlaps the blocking temperatures of the other components, and therefore great circle analysis was used to calculate the magnetic direction. The B component is not present in either sites wm9 or wvs.

In Figure 5 the site means used for calculating the mean direction of component B are plotted both in situ and tilt corrected. The site means and their associated statistical parameters are also listed in Table 1. The directions disperse upon application of the bedding correction, thus failing the fold test [McElhinny, 1964]. The B component is therefore a secondary partial remagnetization, with mean D=167°, I=5°.

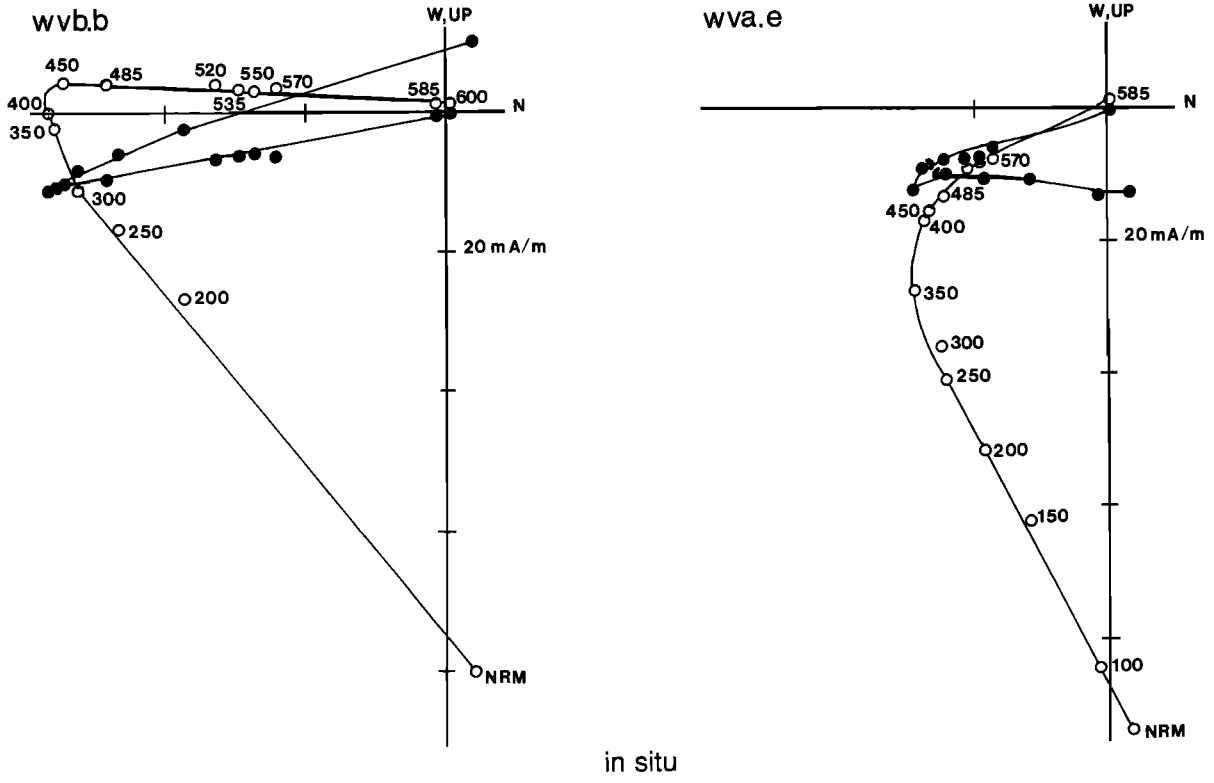


Fig. 4. In situ orthogonal vector diagrams for thermal demagnetizations of samples in which the C and B components of magnetization are found. The successive treatment steps, in °C, are indicated in each diagram. In this and subsequent figures, solid circles represent projections on the horizontal plane and open circles represent projections on the vertical plane. The letters and numbers before the period indicate the site name and after the period indicate the sample number.

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

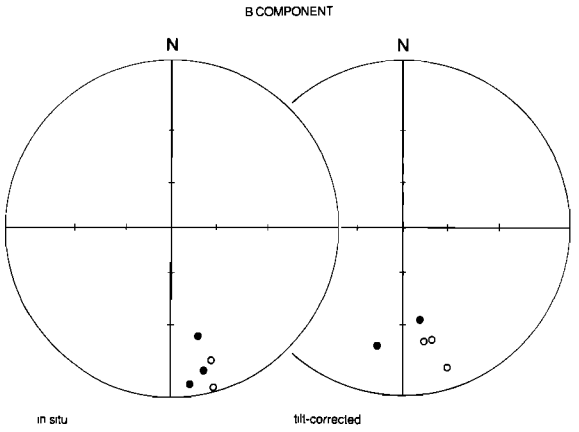


Fig. 5. Equal angle projections of the B component site means both in situ and tilt corrected. The site mean directions and their associated statistical parameters are listed in Table 1. Solid circles represent projections on the lower hemisphere (positive inclinations) and open circles represent projections on the upper hemisphere (negative inclinations).

A minimum of 20 samples from 4 sites (wvs, wm9, wmc, and wva) clearly exhibit the third component, or A component, of magnetization. In several other samples a hint of the third component can be seen in the orthogonal vector diagrams but cannot be defined precisely enough to be included in the mean direction calculations. In sites wvs and wm9 the A direction can be seen as a linear segment on the orthogonal vector diagrams as northwesterly and intermediate down in situ (Figure 6) and as easterly and intermediate down when tilt corrected. In these two sites the B component is not present. When the B component is present, however, the A component can only be defined in samples that clearly show two distinct great circle segments (an indication that three components of magnetization are present). The blocking temperatures of components A, B, and C must therefore overlap, resulting in curvilinear demagnetization diagrams when all three components are present in a sample (Figure 7). Thus in sites wva and wmc, great circle analysis was applied to the high-temperature great circle paths, usually between 350°C and 585°C. A direction antipodal to the typical tilt-corrected easterly direction is indicated by the great circle paths for samples from site wva (Figure 7).

Using the method of Bailey and Halls [1984], the directional and great circle data were combined to yield the mean direction for the A component. The sample directional data and poles to great circle planes for the high-temperature component are plotted in Figure 8, both in situ and tilt corrected. The individual sample directions and pole positions are listed in Table 2. Individual samples were used in calculating the mean direction rather than site means because of the small number of samples available for calculating the site means. The in situ linear directions cluster tightly around a mean A direction, D=310°, I=47°, $\alpha 95=7.2°$. The in situ linear sample directions and great circle poles have been plotted separately (Figures 8a and 8b) to illustrate the ambiguity of the Bailey and Halls analysis prior to tilt correction. The assumption of Bailey and Halls

Table 1. Summary of Secondary Site Mean Paleomagnetic Directions (B Direction)

Site	S/D	n/N	D/I In Situ	D/I Tilt Corrected	k	α 95
wm8	022/68 E	4/7	166/23	165/-21	17.1	22.8
wva	185/86 W	8/8	167/8	192/19	54.2	15.7
wvb	224/51 N	5/9	163/-11	169/32	50.7	10.9
wmc	154/40 W	10/10	165/-1	162/-8	33.8	14.4
wvs	020/90 E	0/6	---	---	---	---
wm9	020/90 E	0/7	---	---	---	---
wvn	081/74 S	0/8	---	---	---	---
wvo	096/61 S	0/7	---	---	---	---
wvr	185/71 W	0/12	---	---	---	---
wv4	128/36 S	6/10	173/4	169/-21	24.9	13.7
wv3	128/36 S	0/11	---	---	---	---

S/D is the strike and dip of the beds used for the tilt correction; n/N is the ratio of the number of samples in which a direction was isolated to the number of samples demagnetized; D/I is the declination and inclination of the paleomagnetic direction, both in situ and tilt corrected; k and $\alpha 95$ are the associated statistical parameters; sites wvs and wm9 did not reveal the B direction; sites wvn, wvo, and wvr had only unstable magnetizations, and the incoherent directions of site wv3 are attributed to a lightning strike.

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

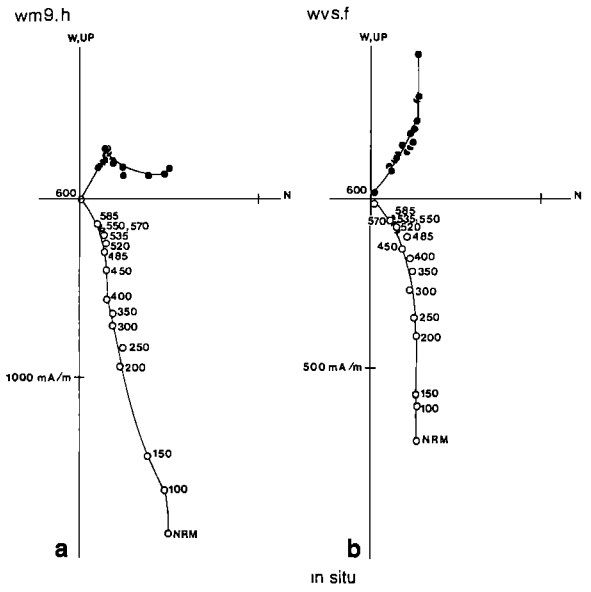


Fig. 6. In situ orthogonal vector diagrams showing thermal demagnetizations of samples that do not contain the B component of magnetization but show the A component at high temperatures. The same plotting conventions as in Figure 4 are used.

analysis is that the great circle poles should form a girdle distribution whose pole corresponds to the mean direction derived from linear directions. The advantage of the Bailey and Halls method is that the number of samples used in an analysis can be increased by combining great circle and directional data. The method, however, is only valid when samples carrying the same magnetic component are being combined. In Figure 8b the great circle poles form a girdle distribution that yields a mean direction that is similar to the B component of magnetization (southerly and shallow) and not at all similar to the northwesterly direction of the A component in Figure 8a. As discussed previously, the samples giving linear directions (Figure 8a) do not contain the B component. Thus, in situ, the directional and great circle data represent different components of magnetization, and therefore they cannot be combined to yield a mean direction that has any significance.

After tilt correction the linear directions and great circle poles appear to define the same component of magnetization (Figure 8c). When the linear directions are tilt corrected, the mean direction is easterly and intermediate down. For the great circle analysis to be successful in defining the third component (presumably the A component) the great circle poles themselves should be distributed along a girdle: the

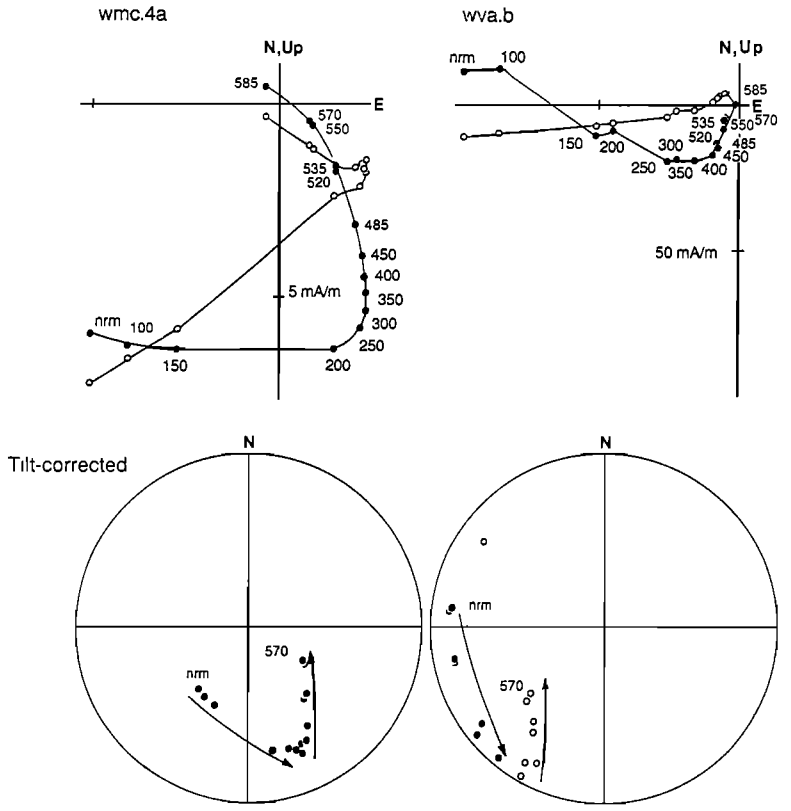


Fig. 7. Tilt-corrected orthogonal vector diagrams for thermally demagnetized samples, with corresponding equal angle projections of demagnetization steps also after tilt correction. All three components of magnetization (A, B, and C) are present in these samples. The samples show antipodal trends for the high-temperature great circle segments. The same plotting conventions as in Figures 4 and 5 are used.

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

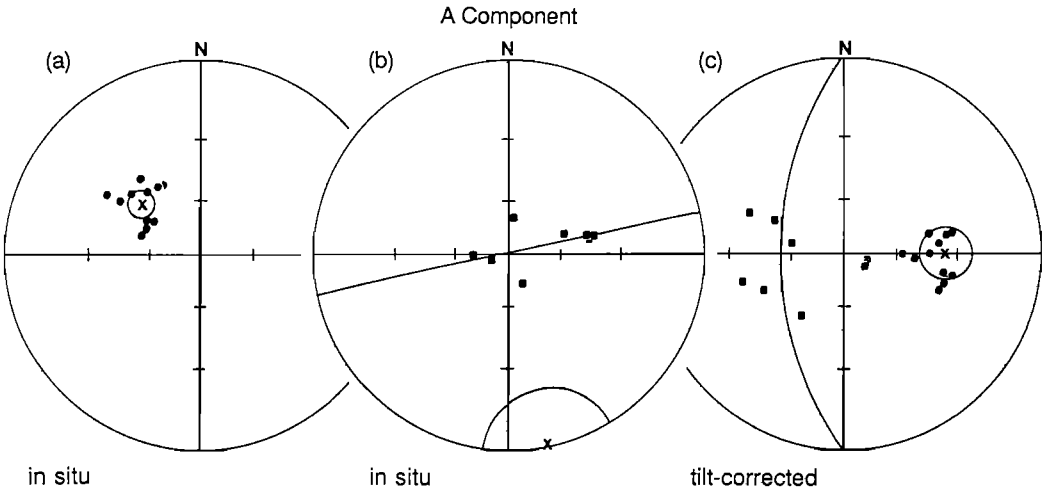


Fig. 8. (a) Equal angle projection of in situ sample directions representing the A component of magnetization in samples that do not contain the B component. The x is the mean direction for the A component in situ with its associated α95. (b) Equal angle projection of in situ poles to high-temperature great circle planes for samples with curvilinear orthogonal vector diagrams. The x is the mean direction calculated from the great circle intersection with its associated α95. The great circle plotted is orthogonal to the mean direction. Note that the mean is similar to the B component and dissimilar to the in situ A direction plotted in Figure 8a. (c) Equal angle projection of the combined tilt-corrected sample directions (circles) and great circle poles (squares), after the method of Bailey and Halls [1984]. The x is the mean direction for the A component calculated from the combined data (see Table 3). The associated α95 is shown as a circle around the mean (x). The great circle plotted is orthogonal to the mean direction.

pole to the best-fitting great circle through this girdle is then the desired direction. In tilt-corrected coordinates this component, if accurately represented by the girdle, would be the prefolding magnetization. We are hard-pressed, however, to define the girdle elongation of the distribution of poles (squares in Figure 8c) by themselves. On the other hand, many great circle poles fall near the girdle distribution to be expected from the linear direction (Figure 8c). Thus it appears that these poles and the tilt-corrected stable endpoint directions, when combined in a Bailey and Halls analysis, yield a satisfactory mean direction (Table 3), and we argue that both represent the tilt-corrected A direction. The mean direction in situ is thus D=310°, I=47°, based on the directional data alone, and tilt corrected it is D=90°, I=36° based on the combined directional and great circle data. A fold test, as defined by McElhinny [1964], is not possible for these data. In the next section, however, we shall argue that the tilt-corrected easterly direction is indeed prefolding.

In determining whether the A component of magnetization is primary or secondary, it is important first to discuss the magnetic carriers in these samples. Isothermal remanent magnetization (IRM) experiments were performed on representative samples of the volcanics. The samples were subjected to progressively higher magnetic fields up to a maximum field strength of 1.4 T. Representative IRM acquisition curves for the volcanics are shown in Figure 9. The three diagrams show samples that contain (a) components B and C only, (b) components A and C only and (c) components A, B, and C. All the samples are saturated by 0.3 T, the theoretical saturation point of magnetite [McElhinny, 1973], indicating that only magnetite is present. In addition, the blocking temperatures of all the samples are consistently near 585°C, the Curie temperature of pure magnetite [McElhinny, 1973]. These data support the conclusion that magnetite is likely to be the magnetic carrier of all three components of magnetization.

Reflected-light microscopy indicates that there are two size ranges of iron oxides in the Stacyville volcanics. In some sites (wvs and wm8) the oxides are abundant (up to 10%) with a restricted grain size of 1-5 μm. The grains are euhedral to subhedral, roughly equant shaped, and are distributed uniformly throughout the matrix. These grains are believed to be magnetite, although minor hematite overgrowths can be seen on some grains. These small magnetite grains are less abundant in sites such as wv4 where the overall remanent magnetization intensity is lower. In site wmc, large euhedral magnetite grains 100-200 μm in size are present, with some minor alteration by hematite occurring along their margins. Magnetite appears to be the predominant iron oxide in the volcanics, although it is found in varying size ranges and abundances.

Interpretation of Paleomagnetic Results

The paleopoles corresponding to the B and A components of magnetization are listed in Table 3.

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Sample Directions and Poles to Great Circles (A Direction)

Site	S/D	Sample	Directions		Poles	
			D/I In Situ	D/I Tilt Corrected	D/I In Situ	D/I Tilt Corrected
wva	185/86 W	b			070/56	291/34
		c			077/45	293/46
wmc	154/40 W	3			008/69	278/58
		4			269/70	254/32
		5			250/80	245/40
		6			154/71	216/46
		7			080/44	132/77
		9			078/42	119/78
		10			079/43	125/77
wvs	020/90 E	a	302/54	101/35		
		b	331/47	079/31		
		e	327/46	080/34		
		f	302/31	091/57		
		g	310/40	090/43		
wm9	020/90 E	a	321/38	077/42		
		b	295/56	107/34		
		c	288/55	111/35		
		e	305/58	101/31		
		f	319/45	084/38		
		h	303/38	094/50		

S/D is the strike and dip of the beds used for the tilt correction; D/I for the directions corresponds to the declination and inclination of the linear segments of magnetic vectors, both in situ and tilt corrected; D/I for the poles corresponds to the declination and inclination of the poles to the great circles for curvilinear demagnetization trajectories, both in situ and tilt corrected.

The B direction yields a paleopole of 40°N, 129°E that falls on the Upper Carboniferous section of the North American apparent polar wander path. The pole position therefore suggests that partial remagnetization of the volcanics occurred during the Carboniferous. Overprints of Carboniferous age have been recognized extensively throughout the Appalachians [e.g., Miller and Kent, 1988].

If the northwesterly magnetization of the A component was acquired after the rocks were folded in the Devonian (post-F2), the direction of magnetization should coincide with a Devonian or younger direction for North America, since the volcanics were contiguous with North America by that time. The in situ direction for the A component of magnetization (D=310°, I=47°), however, is not similar to any other directions known for North America. Therefore the A component of magnetization is not believed to be a post-F2 remagnetization. Additionally, the fact that the directional and great circle data are consistent when tilt corrected, but are not when in situ, indicates that the tilt-corrected easterly direction (D=90°, I=36°) is a pre-F2 magnetization. We recall that the F1 (recumbent) folding did not affect the tilt of the Stacyville volcanics. The inclination (and hence the paleolatitude) is therefore representative of the field at the time the magnetization was acquired. One possibility is that the magnetization is pre-F2 but secondary (i.e., a remagnetization during the Late Ordovician or Siluro-Devonian). Alternatively, the magnetization could be primary.

The paleopole corresponding to the tilt-corrected direction of component A, 14°S, 188°E (see Table 3), is dissimilar to the set of Ordovician poles for North America (Figure 10). The area would have to have been rotated approximately 25°-80° counterclockwise relative to North America in order to account for the differences between the pole from this study and the individual Ordovician paleopoles for the craton. Note that although the pole positions are quite varied, they are approximately equidistant from the site in Maine. Thus the distribution of the poles indicates that North America stayed at roughly the same paleolatitude throughout the Ordovician.

From the inclination data it is possible to calculate the paleolatitude at which the volcanics acquired their magnetization. The mean inclination of 36° corresponds to a paleolatitude of 20°. Given the pre-Devonian paleogeographic setting of the North American margin (see discussion below), it is most likely that the paleolatitudinal position of the volcanics was in the southern hemisphere.

For a comparison between the paleolatitudes expected for North America in the Ordovician and the paleolatitude of this study area we have calculated the expected paleolatitudinal positions for North

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3. Summary of Mean Directions and Paleopoles

Component	n/N	s/S	D/I In Situ	D/I Tilt Corrected	k1	k2	k1/k2	Paleopole
B	5/11		167/5	(171/-1)	38.8	9.3	4.2	40 N, 129 E
A		20/95	(310/47)	090/36	---	12.4	---	14 S, 188 E

The ratio of the number of site means used in calculating the mean to the total number of sites sampled is n/N; the ratio of the number of samples used in calculating the mean to the total number of samples demagnetized is s/S; the mean declinations and inclinations are D/I, both in situ and tilt corrected; the precision parameters before and after tilt correction are k1 and k2, respectively.

America from the Ordovician paleopoles in Figure 10. The paleolatitudes were calculated for a locality at 46°N, 292°E, the present coordinates of the study area (Table 4). In Figure 11 the predicted Ordovician paleolatitudinal positions of North America are plotted versus time. The horizontal error bars correspond to the uncertainty in the age of the magnetization, and the vertical error bars correspond to the A95, or 95% confidence circle around the mean paleopole. The paleolatitudinal position for the Stacyville volcanics (20°S) is also plotted with its associated error limits. The vertical error bar for the volcanics, however, corresponds to the semiaxis (dp) of the 95% confidence oval around a pole calculated from the overall mean direction. An A95 could not be calculated for this study since individual site means and their corresponding poles could not be calculated.

The paleolatitude of the Stacyville volcanics (20°S) is slightly south of the paleolatitude predicted for North America from the Middle Ordovician paleopoles (about 15°S) (Figure 11). If, however, the possibility of a Late Ordovician remagnetization is considered, the paleolatitude of the Stacyville volcanics would be indistinguishable from that of North America, since North America moved to a slightly higher paleolatitude in the Late Ordovician (Figure 11). We can exclude the possibility of a Siluro-Devonian remagnetization, because at that time North America was at higher paleolatitudes (25°-30°S) [Van der Voo, 1988] than the volcanics. Thus we are left with either (1) a primary age for the magnetization or (2) a remagnetization of Late Ordovician age. We favor the interpretation of a primary magnetization based on both the IRM and blocking temperature data, which suggest that magnetite is the magnetic carrier.

DISCUSSION

By combining the geologic and paleomagnetic data a model for the tectonic evolution of the Lunksoos area can be formulated. Before discussing the details of the Lunksoos area, however, it is necessary to

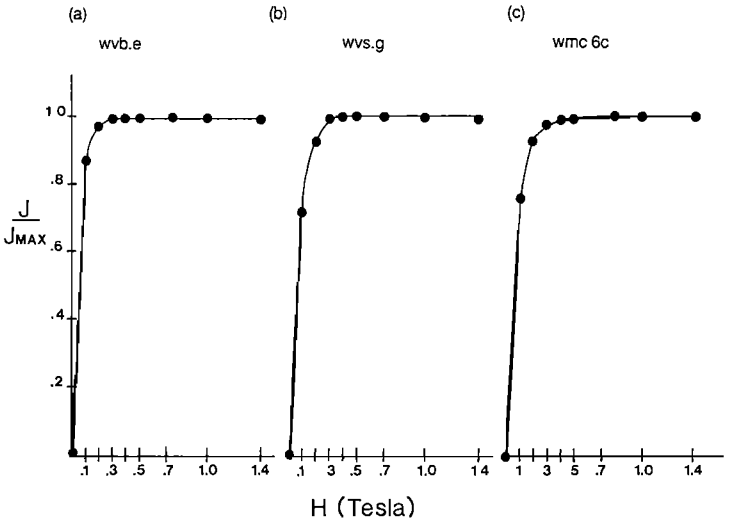


Fig. 9. Typical isothermal remanent magnetization acquisition curves for a sample that contains (a) the B and C components of magnetization, (b) the A and C components of magnetization, and (c) the A, B, and C components of magnetization. H is the intensity of the magnetic field applied to samples at room temperature. J/Jmax is the ratio of the remanence intensity for each step to the maximum remanence intensity.

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

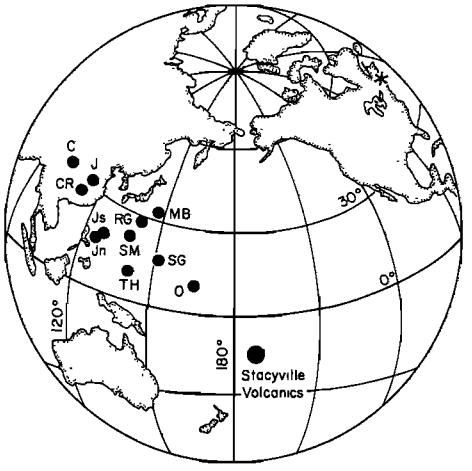


Fig. 10. Ordovician paleomagnetic poles for North America (Table 4) and for this study (A component, Table 3). The labels correspond to those in Table 4, and the star shows the approximate position of the sampling area in Maine.

understand the framework of the surrounding areas during the Ordovician. A fundamental question is the paleoposition and depositional environment of the Cambro-Ordovician rock units within the terrane and the nature of their contacts. The current opinion is that one or more ocean basins had opened widely by the Ordovician, bordered by two major continental blocks, Laurentia (North America) and Armorica-Gondwana [Osberg, 1978; Williams 1979; Cocks and Fortey, 1982; van der Pluijm and van Staal, 1988]. The paleoposition of North America is fairly well constrained by paleomagnetic and paleontologic data for most of the Paleozoic [Van der Voo, 1988]. The North American margin was oriented roughly (north)east-(south)west at near equatorial positions. During the Early and Middle Ordovician the North American margin in the vicinity of the study area in northern Maine was at roughly 15°S. In the Late Ordovician, North America moved south to a paleolatitude of 20°-22°S [Van der Voo, 1988].

It is generally believed that Avalon and Gondwana were together during the Early Paleozoic [Johnson and Van der Voo, 1986]. Avalon, which is thought to have been part of the Armorican block, was on the southern margin of Iapetus. Since this study concerns the area between the North American and Avalonian margins, our discussion will involve only the relative movements of Avalon/Armorica with respect to North America. The paleoposition of Avalon is constrained by paleomagnetic and paleontologic data to be at relatively high southerly paleolatitudes during the Ordovician [Van der Voo, 1988]. There is some uncertainty, however, over its exact position, given that there are only two results from Avalon for the Early and Middle Ordovician: the Nahant Gabbro [Weisse et al., 1985] and the Cape Breton Granites [Rao et al., 1981]. Both of these results are from intrusive rocks, so structural control is lacking. The paleolatitudes for Avalon derived from these studies are very high (60°-70°S). A recent study of Middle

TABLE 4. Ordovician Paleolatitudes for North America

Formation	Age	Paleopole	A95	dp,dm	Paleolatitude	Reference
Oneota (O)	Ol	10 N, 166 E	7.0	5.7, 11.4	16	[Jackson and Van der Voo, 1985]
St. George (SG)	Ol	17 N, 152 E	3.4 (c)	2.8, 4.9	17	[Deutsch and Prasad, 1987]
Moccasin, Bays (MB)	Om	33 N, 147 E	5.8	---	5	[Watts and Van der Voo, 1979]
Chapman Ridge (CR)	Om	27 N, 112 E	15.0	---	17	[Watts and Van der Voo, 1979]
Steel Mountain (SM)	451±35	22 N, 139 E	---	8, 14	---	[Murthy and Rao, 1976]
		20 N, 141 E (c)	14.7 (c)	---	19	
Juniata (J)	Ou	32 N, 114 E	4.8	3.1, 5.5	12	[Van der Voo and French, 1977]
Juniata North (Jn)	Ou	21 N, 105 E	13.0	---	---	[Miller and Kent, 1989]
		17 N, 126 E (r)	13.0	---	26	
Juniata South (Js)	Ou	19 N, 128 E	21.0	---	23	[Miller and Kent, 1989]
Cordova (C)	447±8	31 N, 102 E	---	6.5, 12	---	[Dunlop and Stirling, 1985;
		30 N, 102 E (c)	8.6 (c)	---	13	Lopez-Martinez and York, 1983]
Ringgold Gap (RG)	Ou/Sl	28 N, 142 E	7.3	---	11	[Morrison and Ellwood, 1986]
Table Head (TH)	Ou/S	11 N, 142 E	---	11, 9	---	[Hodych, 1989]
		10 N, 142 E (c)	10.0 (c)	---	28	
This study	Om(?)	14 S, 188 E	---	8.4,14.4	20	

The age corresponds to the age of remanent magnetization; A95 is the circle of 95% confidence about the pole; dp and dm are the semiaxes of the 95% confidence oval around the pole; (c) indicates that we have recalculated the pole or A95 from published site means; (r) indicates that for the northern limb of the Pennsylvania salient, the pole has been rotated about a pivot located at 41 N, 77 W, angle 23 degrees counterclockwise, following Kent [1988]; the paleolatitudes are determined from the published data for a site at 46 N, 292 E (the latitude and longitude of the sampling site for this study).

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

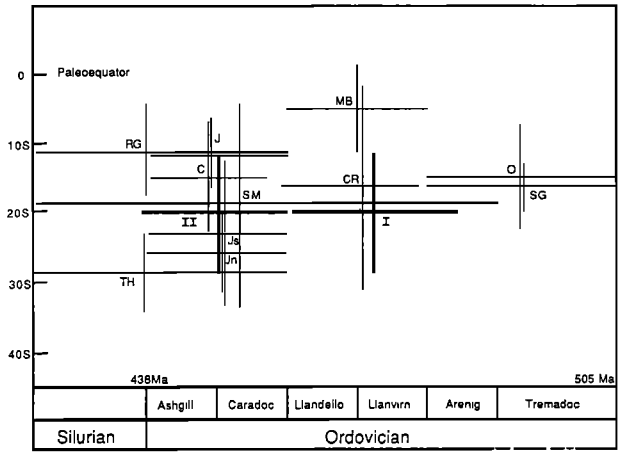


Fig. 11. Ordovician paleolatitudes versus age for the Maine study locality, calculated from the North American poles; paleolatitudes are listed in Table 4. Horizontal error bars correspond to uncertainty in the age of the magnetization. Vertical error bars correspond to the A95 of the pole, or the uncertainty of the paleolatitude. The paleolatitudinal position for this study is also plotted. Note that the vertical error for this study corresponds to dp, the semiaxis of the 95% confidence oval about the pole. The I indicates the position of the volcanics if the magnetization is primary and the II indicates the position if the magnetization is secondary.

Ordovician volcanics from Wales yields a more moderate paleolatitude of 55°S [McCabe and Channell, 1988]. There is no consensus whether southern Britain was part of the Armorican block, but most Appalachian geologists [e.g., Rast and Skehan, 1983; Neuman, 1984] agree that Avalon and southern Britain show many similarities during the Ordovician. Thus a more reliable estimate for the position of Avalon in the Middle Ordovician could perhaps be 50°-55°S. For the Late Ordovician-Early Silurian there is a result from Nova Scotia for the Dunn Point Formation [Van der Voo and Johnson, 1985] that yields a well-constrained paleolatitude of 42°S. Thus Avalon appears to be moving northward during the Ordovician. We conclude that the paleolatitude of Avalon can best be estimated as having been 50°S or higher during the Middle Ordovician, decreasing to about 40°S by the latest Ordovician.

By knowing the paleopositions of North America and Avalon, a reference frame for the intervening area is established, allowing a discussion of the Lunksoos area in terms of its relations to the North American and Avalonian crustal blocks. The unit for which there is the least amount of information is the Cambrian Grand Pitch Formation. The graded beds of quartzites and shales are indicative of a slope-rise sequence. No fossils have been found that are distinctive of a specific faunal province, however [see Lindholm and Casey, 1989]. Our paleomagnetic study of the Grand Pitch Formation was not successful, and therefore its paleoposition remains unconstrained. The Grand Pitch Formation could have been deposited marginal to North America or Avalon. Additionally, it could have been deposited on either side of an unidentified crustal block (microcontinent) within Iapetus. It has been suggested that the Gander Group rocks of Newfoundland and New Brunswick are at least in part equivalent to the Grand Pitch Formation [e.g., Rast et al., 1976]. A Gander zone affinity for the Grand Pitch would indicate that it was formed on the southern margin of the Ordovician Iapetus ocean. We believe that a position close to Avalon, at least during the Cambrian and Early Ordovician, best accommodates the data that will be discussed next.

The overlying Shin Brook Formation consists of tuffs and tuffaceous sandstones that are relatively shallow water deposits. Brachiopods found in the sandstones belong to the Celtic biogeographic province [Neuman, 1984]. The Celtic province is defined in Anglesey by a reference assemblage of brachiopods that are inferred to have occupied intermediate to high latitudes [Cocks and Fortey, 1982; Neuman, 1984]. The high percentage of endemic genera (46%) and the geology associated with the Celtic fauna suggest a peri-insular setting that lay adjacent to the Armorican continental margin [Neuman, 1984]. Consequently, the Shin Brook Formation is believed to have formed at intermediate to high paleolatitudes near the Armorican continental margin.

The conglomerate at the base of the Early Ordovician Shin Brook Formation contains clasts of slate and quartzite presumably derived from the Grand Pitch Formation. This field relationship indicates that the Shin Brook was deposited on top of the Grand Pitch. The position of the underlying Grand Pitch is therefore constrained to be on the southern side of Iapetus, probably near the Armorican margin during the deposition of the Shin Brook in the Early Ordovician.

The paleolatitudinal position of the Middle (?) Ordovician Stacyville volcanics has been determined in this study. The paleomagnetic data indicates that the volcanics acquired their magnetization at a relatively shallow paleolatitude of 20°S during either the Middle or Late Ordovician. This position is clearly distinct from that inferred for the Shin Brook Formation, with Avalon at intermediate to high paleolatitudes during the Middle Ordovician (50°-55°S) and even in the Late Ordovician at paleolatitudes of 40° or higher [Van der Voo and Johnson, 1985]. Thus the volcanics could not have been deposited directly on top of the Shin Brook Formation, and their current relationship must be tectonic. The timing of the emplacement of the volcanics over the Grand Pitch-Shin Brook sequence is constrained by the intrusion of the Rockabema Diorite to have occurred before the late Early Silurian. The age of this tectonic movement coincides with the inferred age of the F1 fold generation, supporting the conclusion of Late

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

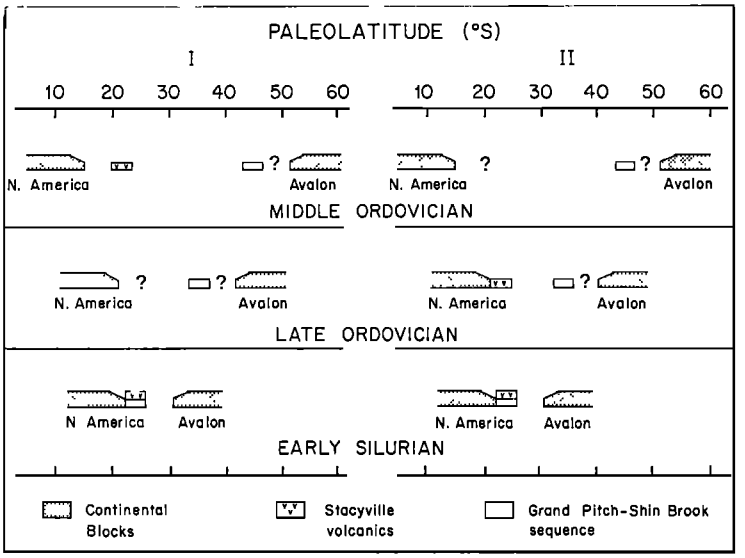


Fig. 12. Schematic diagrams showing the paleolatitudinal positions of North America, the Stacyville volcanics, the Grand Pitch-Shin Brook sequence, and Avalon for the two possible ages of magnetization of the Stacyville volcanics. Positions are based on paleomagnetic, paleontologic, and geologic data and are shown for Middle Ordovician, Late Ordovician, and Early Silurian. Option I shows the position of the volcanics if the magnetization is primary and Option II shows the position if the magnetization is secondary. See text for discussion.

Ordovician to Early Silurian deformation.

Alternatively, the contact between the Stacyville volcanics and the Grand Pitch and Shin Brook Formations could be stratigraphic if the underlying units were not deposited at southerly latitudes, but rather were deposited marginal to North America. Such an interpretation, however, is not supported by the faunal provinciality arguments for a central-to-southern Iapetus location of the Shin Brook Formation [Neuman, 1984].

The timing of the emplacement is particularly intriguing. Generally, two main periods of deformation have been reported in Maine [Osberg, 1978; Williams and Hatcher, 1983]. The Taconic Orogeny is thought to have occurred during the Middle Ordovician and the Acadian Orogeny during the Devonian. The Early Silurian event of this study is therefore neither Taconic nor Acadian by these definitions. The timing of this event implies one of the following: (1) the Taconic Orogeny lasted longer than previously reported in Maine and continued throughout the Ordovician into the Early Silurian, (2) the Taconic Orogeny, as previously defined, occurred in the Middle Ordovician and was followed by another tectonic event in the Late Ordovician-Early Silurian that predated the Acadian Orogeny, (3) the Acadian event started prior to Devonian times and continued into the Devonian.

TECTONIC INTERPRETATION

The data from this study and previous studies allow a tectonic model for the Lunksoos area to be proposed. Figure 12 contains two diagrams showing the positions of North America, the Stacyville volcanics, the Grand Pitch-Shin Brook sequence, and Avalon from the Middle Ordovician to the Early Silurian, based on the two possible ages of magnetization of the Stacyville volcanics, that is, Middle and Late Ordovician, respectively. The paleolatitudinal positions are derived from the data described in the previous sections. During the Middle Ordovician, North America was at a paleolatitude of 15°S and Avalon at 50°-55°S. The Grand Pitch and Shin Brook formations had already been deposited near the Avalonian continental margin and were therefore at a paleolatitude no higher than Avalon, perhaps 45°-50°S. The paleomagnetic data for the Stacyville volcanics yields a paleolatitude of 20°S during the Middle Ordovician if the magnetization is primary. The position of the volcanics is then unconstrained during the Late Ordovician. If the magnetization is secondary, however, the volcanics would have been at a paleolatitude of 20°S during the Late Ordovician, and their position during the Middle Ordovician would be unconstrained by this data. During the Late Ordovician and Early Silurian, North America moved south and Avalon moved north [Van der Voo, 1988]. As North America and Avalon converged, the volcanics were structurally emplaced onto the Grand Pitch-Shin Brook sequence no later than the late Early Silurian, forming the Lunksoos composite terrane. The timing and nature of the tectonic emplacement would be the same regardless of the age of magnetization.

CONCLUSIONS

The results of our study show that the Stacyville volcanics were located near the North American

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

margin and that they were subsequently emplaced over the Grand Pitch and Shin Brook formations. If the arguments that the Grand Pitch-Shin Brook sequence was located near Avalon are valid, the two rock sequences (Stacyville and Grand Pitch-Shin Brook) must have been separated by at least 25°-30° of latitude in Ordovician times. Between the Middle Ordovician when the volcanics were formed and the Early Silurian when the Rockabema Diorite was intruded the volcanics must have been structurally emplaced into their present position.

The combined geologic and paleomagnetic data provide the basis for a tectonic model of the Lunksoos terrane, although the applicability of this tectonic scenario to other terranes in Maine or elsewhere in the northern Appalachians may be limited. Further study of the timing and cause of deformation events and the paleogeographic setting of the terranes in Maine needs to be completed before a comprehensive model can be presented for the tectonic evolution of this part of the northern Appalachians.

Acknowledgments. We thank Bob Neuman for the time he spent introducing us to the geology of the Shin Pond area. We also thank Bill Forbes for his generous assistance whenever we were in Maine. This material is based upon work supported under a National Science Foundation Graduate Fellowship to the first author. Fieldwork for this study was supported by a Grant-in-Aid of Research from Sigma Xi, a Geological Society of America Research Grant, the Scott Turner Fund of the Department of Geological Sciences, the University of Michigan, and the Division of Earth Sciences, National Science Foundation grant EAR 86-12469. Thoughtful reviews were given by Doug Rankin, Hank Williams, and an anonymous reviewer.

REFERENCES

Ayuso, R.A., Lead-isotopic evidence for distinct sources of granite and for distinct basements in the northern Appalachians, Maine, *Geology, 14*, 322-325, 1986.

Bailey, R.C., and H.C. Halls, Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data, *J. Geophys., 54*, 174-182, 1984.

Bird, J.M., and J.F. Dewey, Lithospheric plate-continental margin tectonics and the evolution of the Appalachian orogen, *Geol. Soc. Am. Bull., 81*, 1031-1060, 1970.

Cocks, L.R.M., and R.A. Fortey, Faunal evidence for oceanic separations in the Palaeozoic of Britain, *J. Geol. Soc. London, 139*, 465-478, 1982.

Deutsch, E.R., and J.N. Prasad, Ordovician paleomagnetic results from the St. George and Table Head carbonates of western Newfoundland, *Can. J. Earth Sci., 24*, 1785-1796, 1987.

Dunlop, D.J., and J.M. Stirling, Post-tectonic magnetizations from the Cordova gabbro, Ontario and Palaeozoic reactivation in the Grenville province, *Geophys. J. R. Astron. Soc., 81*, 521-550, 1985.

Halls, H.C., A least-squares method to find a remanence direction from converging remagnetization circles, *Geophys. J. R. Astron. Soc., 45*, 297-304, 1976.

Hodych, J.P., Limestones of western Newfoundland that magnetized before Devonian folding but after Middle Ordovician lithification, *Geophys. Res. Lett., 16*, 93-96, 1989.

Hynes, A., Magmatic affinity of Ordovician volcanic rocks in northern Maine, and their tectonic significance, *Am. J. Sci., 276*, 1208-1224, 1976.

Hynes, A., On the tectonic setting of Ordovician volcanic rocks from northern Maine, *Am. J. Sci., 281*, 545-552, 1981.

Jackson, M., and R. Van der Voo, A lower Ordovician paleomagnetic pole from the Oneota Dolomite, Upper Mississippi River Valley, *J. Geophys. Res., 90*, 10,449-10,461, 1985.

Johnson, R.J.E., and R. Van der Voo, Middle Cambrian paleomagnetism of the Avalon terrane in Cape Breton Island, Nova Scotia, *Tectonics, 4*, 629-651, 1985.

Johnson, R.J.E., and R. Van der Voo, Paleomagnetism of the Late Precambrian Fourchu Group, Cape Breton Island, Nova Scotia, *Can. J. Earth Sci., 23*, 1673-1685, 1986.

Kent, D.V., Further paleomagnetic evidence for oroclinal rotation in the Central Folded Appalachians from the Bloomsburg and the Mauch Chunk Formations, *Tectonics, 7*, 749-759, 1988.

Keppie, J.D., The Appalachian collage, in *The Caledonide Orogen-Scandinavia and Related Areas*, edited by D.G. Gee and B.A. Sturt, pp. 1217-1226, John Wiley and Sons, New York, 1985.

Kirschvink, J.L., The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. Astron. Soc., 62*, 699-718, 1980.

Lindholm, R.M., and J.F. Casey, Regional significance of the Blow Me Down Brook Formation, western Newfoundland: New fossil evidence for an Early Cambrian age, *Geol. Soc. Am. Bull., 100*, 1-13, 1989.

Lopez-Martinez, M., and D. York, Further thermochronometric unravelling of the age and palaeomagnetic record of the southwest Grenville Province, *Can. J. Earth Sci., 20*, 953-960, 1983.

Ludman, A., Timing of terrane accretion in eastern and east-central Maine, *Geology, 14*, 411-414, 1986.

McCabe, C., and J.E.T. Channell, Paleomagnetism of the Middle Ordovician Stapely Volcanic Group, Welsh Borderlands (abstract), *EOS, 69*, 337, 1988.

McElhinny, M.W., Statistical significance of the fold test in paleomagnetism, *Geophys. J. R. Astron. Soc., 8*, 125-134, 1964.

McElhinny, M.W., *Palaeomagnetism and Plate Tectonics*, 358 pp., Cambridge University Press, London, 1973.

Miller, J.D., and D.V. Kent, Regional trends in the timing of Alleghanian remagnetization in the Appalachians, *Geology, 16*, 588-591, 1988.

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Miller, J.D., and D.V. Kent, Paleomagnetism of Upper Ordovician Juniata Formation of the central Appalachians revisited again, *J. Geophys. Res., 94*, 1843-1849, 1989.

Morrison, J., and B.B. Ellwood, Paleomagnetism of Silurian-Ordovician sediments from the Valley and Ridge province, northwest Georgia, *Geophys. Res. Lett., 13*, 189-192, 1986.

Murthy, G.S., and K.V. Rao, Paleomagnetism of Steel Mountain and Indian Head Anorthosites from western Newfoundland, *Can. J. Earth Sci., 13*, 75-83, 1976.

Neuman, R.B., The Grand Pitch Formation: New name for the Grand Falls Formation (Cambrian?) in northeastern Maine, *Am. J. Sci., 260*, 794-797, 1962.

Neuman, R.B., Fossils in Ordovician tuffs, northeastern Maine (with a section on Trilobita by H.B. Whittington), *U.S. Geol. Surv. Bull., 1181-E*, 38 pp., 1964.

Neuman, R.B., Bedrock geology of the Shin Pond and Stacyville quadrangles, Penobscot County, Maine, *U.S. Geol. Surv. Prof. Paper 524-I*, 37 pp., 1967.

Neuman, R.B., Geology and paleobiology of islands in the Ordovician Iapetus Ocean: Review and implications, *Geol. Soc. Am. Bull., 95*, 1188-1201, 1984.

Osberg, P.H., Synthesis of the geology of the northeastern Appalachians, U.S.A., in *Caledonian-Appalachian Orogeny of the North Atlantic Region, Pap. 78-13*, pp. 137-147, Geological Survey of Canada, Ottawa, 1978.

Rankin, D.W., and R. Hon, Traveler Rhyolite and overlying Trout Valley Formation and the Katahdin Pluton; A record of basin sedimentation and Acadian magmatism, north-central Maine, in *Centennial Field Guide, Northeastern Section*, edited by D.C. Roy, pp. 293-301, Geological Society of America, Boulder, Colo., 1987.

Rao, K.V., M.K. Seguin, and E.R. Deutsch, Paleomagnetism of Siluro-Devonian and Cambrian granitic rocks from the Avalon zone in Cape Breton Island, Nova Scotia, *Can. J. Earth Sci., 18*, 1187-1210, 1981.

Rast, N., and J.W. Skehan, The evolution of the Avalonian plate, *Tectonophysics, 100*, 257-286, 1983.

Rast, N., M.J. Kennedy, and R.F. Blackwood, Comparison of some tectonostratigraphic zones in the Appalachians of Newfoundland and New Brunswick, *Can. J. Earth Sci., 13*, 868-875, 1976.

Rodgers, J., Chronology of tectonic movements in the Appalachian region of eastern North America, *Am. J. Sci., 265*, 408-427, 1967.

Smith, E.S.C., The Cambrian in northern Maine, *Am. J. Sci., 15*, 484-486, 1928.

van der Pluijm, B.A., and C.R. van Staal, Characteristics and evolution of the Central Mobile Belt, Canadian Appalachians, *J. Geol., 96*, 535-547, 1988.

Van der Voo, R., Paleozoic paleogeography of North America, Gondwana, and intervening displaced terranes: Comparisons of paleomagnetism with paleoclimatology and biogeographical patterns, *Geol. Soc. Am. Bull., 100*, 311-324, 1988.

Van der Voo, R., and R.B. French, Paleomagnetism of the Late Ordovician Juniata Formation and the remagnetization hypothesis, *J. Geophys. Res., 82*, 5796-5802, 1977.

Van der Voo, R., and R.J.E. Johnson, Paleomagnetism of the Dunn Point Formation (Nova Scotia): High paleolatitudes for the Avalon terrane in the Late Ordovician, *Geophys. Res. Lett., 12*, 337-340, 1985.

Watts, D.R., and R. Van der Voo, Paleomagnetic results from the Ordovician Moccasin, Bays, and Chapman Ridge Formations of the Valley and Ridge province, eastern Tennessee, *J. Geophys. Res., 84*, 645-655, 1979.

Weisse, P.A., S.E. Haggerty, and L.L. Brown, Paleomagnetism and magnetic mineralogy of the Nahant Gabbro and Tonalite, eastern Massachusetts, *Can. J. Earth Sci., 22*, 1425-1435, 1985.

Wellensiek, M.R., Tectonic History of the Ordovician Weeksboro-Lunksoos Lake Terrane in the Maine Appalachians, M.S. thesis, 47 pp., Univ. of Michigan, Ann Arbor, 1988.

Williams, H., Appalachian Orogen in Canada, *Can. J. Earth Sci., 16*, 792-807, 1979.

Williams, H., and R.D. Hatcher, Jr., Appalachian suspect terranes, in *Contributions to the Tectonics and Geophysics of Mountain Chains*, vol. 158, edited by R.D. Hatcher, Jr., H. Williams, and I. Zietz, pp. 33-53, Memoirs of the Geological Society of America, Boulder, Colo., 1983.

Zartman, R.E., Three decades of geochronologic studies in the New England Appalachians, *Geol. Soc. Am. Bull., 100*, 1168-1180, 1988.

Zen, E., Exotic terranes in the New England Appalachians-limits, candidates and ages: A speculative essay, in *Contributions to the Tectonics and Geophysics of Mountain Chains*, vol. 158, edited by R.D. Hatcher, Jr., H. Williams, and I. Zietz, pp. 55-81, Memoirs of the Geological Society of America, Boulder, Colo., 1983.

Zijderveld, J.D.A., AC demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism*, edited by D.W. Collinson, K.M. Creer, and S.K. Runcorn, pp. 254-286, Elsevier, Amsterdam, The Netherlands, 1967.

R.J.E. Johnson, B.A. van der Pluijm, R. Van der Voo and M.R. Wellensiek, Department of Geological Sciences, 1006 C.C. Little Building, University of Michigan, Ann Arbor, MI 48109.

(Received March 22, 1989;
revised September 22, 1989;
accepted September 25, 1989.)

19449194, 1990, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC009i004p00719 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License